

Critical Review

Updating the Chronic Freshwater Ecotoxicity Database and Biotic Ligand Model for Nickel for Regulatory Applications in Europe

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Abstract: Bioavailability has been taken into account in the regulation of nickel in freshwater ecosystems in Europe for over a decade; during that time a significant amount of new information has become available covering both the sensitivity of aquatic organisms to nickel toxicity and bioavailability normalization. The ecotoxicity database for chronic nickel toxicity to freshwater organisms has been updated and now includes 358 individual data points covering a total of 53 different species, all of which are suitable for bioavailability normalization to different water chemistry conditions. The bioavailability normalization procedure has also been updated to include updates to the bioavailability models that enable more sensitive water chemistry conditions to be covered by the model predictions. The updated database and bioavailability normalization procedure are applicable to more than 95% of regulated European surface water conditions and have been used to calculate site-specific criteria for a variety of different water chemistry scenarios, to provide an indication of how the sensitivity to nickel varies between different water types. The hazardous concentration for 5% of a species (HC5) values for this diverse selection of water types range from 1.6 to 36 $\mu\text{g L}^{-1}$, clearly demonstrating the importance of accounting for nickel bioavailability in freshwaters. This updated database and bioavailability normalization procedure provide a robust basis for the derivation of regulatory thresholds for chronic nickel toxicity in freshwaters such as predicted no-effect concentrations and Environmental Quality Standards and are protective of the results of several mesocosm studies. *Environ Toxicol Chem* 2023;42:566–580. © 2023 The Authors. *Environmental Toxicology and Chemistry* published by Wiley Periodicals LLC on behalf of SETAC.

Keywords: Bioavailability; Water quality criteria; Trace metals

INTRODUCTION

Several abiotic factors influence chronic aquatic nickel (Ni) toxicity. The most important toxicity-modifying factors for Ni in laboratory toxicity tests are pH, hardness (calcium and magnesium), and dissolved organic carbon (DOC; Deleebeeck et al., 2008; Deleebeeck, De Schamphelaere, & Janssen, 2009; Deleebeeck, De Laender, et al., 2009; Santore et al., 2021).

To incorporate the influence of these factors into toxicity predictions, chronic Ni bioavailability normalization models have been developed and validated for different species from three trophic levels. These models incorporate the effects of aqueous speciation and competition of cations such as Ca^{2+} and Mg^{2+} for binding at a theoretical biotic ligand site. Nickel bioavailability normalization models currently exist for the invertebrates *Ceriodaphnia dubia* (De Schamphelaere et al., 2006) and *Daphnia magna* (Deleebeeck et al., 2008), the alga *Pseudokirchneriella subcapitata* (Deleebeeck, De Schamphelaere, & Janssen, 2009), and the fish *Oncorhynchus mykiss* (Deleebeeck, De Schamphelaere, & Janssen, 2007; Deleebeeck, Muysen, et al., 2007). The alga *P. subcapitata* was recently renamed *Raphidocelis subcapitata*, but the former name is used in the present study for consistency with the older studies that report the data we are concerned with. Specific examples of

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Published online 18 January 2023 in Wiley Online Library

(wileyonlinelibrary.com).

DOI: 10.1002/etc.5561

cross-species extrapolation of the Ni bioavailability models have been demonstrated by applying the *O. mykiss* bioavailability model to another fish, that is, *Pimephales promelas* (Deleebeeck, De Schamphelaere, & Janssen, 2007), by applying the *D. magna* bioavailability model to 10 other cladoceran species (Deleebeeck, Muyssen, et al., 2007), and by applying the *P. subcapitata* bioavailability model to 10 green algae species (Deleebeeck, De Laender, et al., 2009b). Schlek et al. (2010) also demonstrated that the *C. dubia* and *D. magna* bioavailability models could predict Ni toxicity to other non-cladoceran invertebrates (the rotifer *Brachionus calyciflorus*, the insect *Chironomus tentans*, and the mollusc *Lymnaea stagnalis*), as well as the aquatic plant *Lemna minor* within a factor of 2. More recently Peters et al. (2018) demonstrated the applicability of similar models to a number of tropical and subtropical species. The application of species-specific bioavailability models to non-model species is referred to as cross-species extrapolation and has been demonstrated to provide reliable results for a variety of representative organisms from all three trophic levels (Peters et al., 2018; Schlek et al., 2010).

In 2008, the European Union Risk Assessment Report (RAR) on Ni (European Commission [EC]) was published. The Ni bioavailability models were applied to take account of the influence of water chemistry on toxicity. The bioavailability normalization process is equivalent to performing all the ecotoxicity tests that are included in the ecotoxicity database in waters matched to the characteristics of the site under consideration. Use of the bioavailability models allows predictions of chronic ecotoxicity endpoints to be made for specific local conditions that are the basis for species sensitivity distributions (SSDs), to derive relevant environmental thresholds. The approaches from the European Union RAR were subsequently adopted as the basis for derivation of the Environmental Quality Standard (EQS) for Ni in European freshwaters under the Water Framework Directive (WFD).

The composite modeling approach that integrates the ecotoxicity database with the species-specific bioavailability models to calculate a bioavailability normalized SSD is referred to as the full biotic ligand model (BLM); for clarity, in the present study the term “species-specific bioavailability models” is used to refer to the individual models for *P. subcapitata*, *D. magna*, *C. dubia*, and *O. mykiss*. The ecotoxicity database and bioavailability normalization procedures underlying the EQS derivation have been published by Nys et al. (2016), and a simplified tool called biomet has been developed to facilitate compliance checking (Peters et al. 2020). However, although these models have been subject to rigorous validation using ecotoxicity tests performed in field-collected natural waters, the models have only been validated for pH values of 8.2 or less and Ca values of 2 mg/L or more. Based on the Forum of European Geological Survey (FOREGS) database (Salminen et al., 2005), 14% of European surface waters have a pH above 8.2, and 6% of the surface waters have a Ca concentration below 2 mg/L. This is important because conditions of high pH and/or low Ca represent conditions of high Ni bioavailability and hence higher ecosystem sensitivity (Deleebeeck, De Laender, et al., 2009; Deleebeeck, De Schamphelaere, & Janssen, 2007; Deleebeeck et al., 2008; Deleebeeck,

De Schamphelaere, & Janssen, 2009; Nys et al., 2016; Peters et al., 2018). Approximately 15% of European surface waters with potentially high Ni bioavailability are outside the boundaries of the bioavailability normalization approach currently used for predicted no-effect concentration (PNEC) and EQS derivation based on regulatory monitoring data (see Merrington et al., 2020), making the applicability of the current thresholds in these waters uncertain.

To address these limitations, two recent studies have demonstrated that the original bioavailability models can be modified for application over a broader range of conditions, to include higher pH (up to pH 8.7; Nys et al., 2016), and also to lower calcium concentrations (down to 0.5 mg/L; Peters et al., 2018) than was possible previously. The low Ca adaptations of this modeling (Peters et al., 2018) followed a validation approach similar to that used for the original models (Schlek et al., 2010) by demonstrating their applicability in laboratory ecotoxicity tests using field-collected natural waters. Peters et al. (2018) also demonstrated that the algal and vertebrate models could be applied to other species from the same trophic level. The incorporation of these high-pH and low-Ca model applications into the PNEC or EQS derivation approach would increase the applicability of these environmental threshold values across European waters (Peters et al., 2022). In addition, the chronic Ni toxicity database used to derive the current Ni EQS is based on a data collation exercise that ended in 2008, whereas several high-quality studies considering chronic Ni toxicity to aquatic organisms have been published since then.

The present study aims to revise the environmental threshold calculation method for Ni by incorporating the most recent available scientific information. This was done by first updating the ecotoxicity database used for the existing Ni PNEC and EQS and combining it with additional studies that allow the bioavailability normalization models to be applied to a wider range of water chemistry conditions (Nys et al., 2016; Peters et al., 2018). The present study also revised the current cross-species extrapolation method to a more evidence-based approach. The approach taken for the averaging of data from multiple studies for the same species was also updated to use an intrinsic sensitivity coefficient for each species rather than normalizing the results for each individual entry in the ecotoxicity database.

The process followed for these updates is shown as a schematic in Figure 1. The updated ecotoxicity database and bioavailability normalization procedure is applied to a variety of water types to calculate site-specific hazardous concentration for 5% of a species (HC5) values for these water chemistry conditions, to give an overview of the range of site-specific HC5 values that can be expected in European surface waters. These updates provide the basis for potential revisions of the existing regulatory thresholds for chronic Ni toxicity in freshwaters, such as the EQS under the WFD and the PNEC value under Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) regulations, recognizing that maintaining unified methods across different regulatory regimes in Europe is desirable.

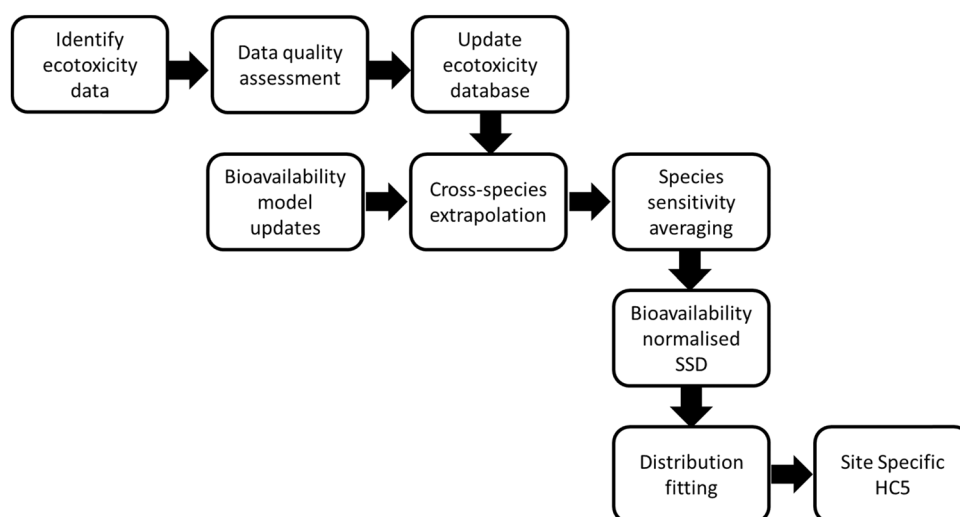


FIGURE 1: Schematic of the process followed for updates to the bioavailability normalization of the updated nickel ecotoxicity database. SSD = species sensitivity distribution; HC5 = hazardous concentration to 50% of the species.

MATERIALS AND METHODS

Compilation of chronic freshwater toxicity database

The European Union RAR for Ni and Ni compounds (EC, 2008) was used as the basis for the compilation of chronic freshwater toxicity data for Ni in the present study. The European Union RAR contains 214 toxicity datapoints retrieved from peer-reviewed literature and published study reports, which were assessed based on relevancy and reliability. The data set including detailed water chemistry has been published in Nys et al. (2016). As a first step toward the compilation of the present data set, the collected 10% effective concentration (EC10) values and no-observed-effect concentrations (NOECs) from the 2008 database were re-evaluated against the most recent available publications, and the values used were updated where appropriate. In addition, if an EC10 had since become available, the NOEC originally used was replaced by the EC10. Second, the literature was searched for newly generated toxicity data since the data collection for the 2008 report (up to 2020). New toxicity data were evaluated for reliability and relevance using guidelines set out in the guidance of the REACH regulatory framework (European Chemicals Agency [ECHA], 2008a) and metal-specific guidance published in the context of the current REACH regulation (ECHA, 2008b). Only toxicity data that passed the quality screening were included in the effects assessment database of Ni.

Reliability was assessed based on a sufficiently described methodology, measured (preferably dissolved Ni, although total Ni was also retained) toxicity data were preferred (nominal toxicity values were rejected), the dose–response relationship was clear, and control performance criteria were acceptable. Both the EC10 and 10% lethal concentration (LC10) and the NOEC values were considered suitable for HC5–50 derivation, but EC10/LC10 values were used when they were available.

The relevant criteria of toxicity data considered the choice of test species, test substance, and test duration. All freshwater

species for which Ni toxicity data are available were included, including species or life stages that have no standardized test procedure. Only Ni-only exposures, based on additions of soluble Ni salts, and using test media representative for the environmental compartment studied (both natural and artificial media), were considered. Given the importance of bioavailability, only data for which sufficient information on water chemistry are reported. The DOC of synthetic test media with no organic matter added was set to 0 mg/L (EC, 2008; Nys et al., 2016). A detailed list of assumptions used for the estimation of missing water chemical parameters is given in the toxicity database (Supporting Information S1). A detailed description of data selection criteria covering the assessment of both reliability and relevance is provided in the Supporting Information S2.

Bioavailability normalization procedure: Integration of the bioavailability models in the full BLM of Ni

The full BLM normalization procedure combines the existing Ni bioavailability models with speciation calculations in the speciation software Windermere Humic Aqueous Model VI (WHAM VI; Tipping, 1998). The same speciation software version and speciation assumptions used in the Ni RAR and for setting the 2011 Ni EQS (EC, 2011) were also applied in the present study. Details on speciation assumptions can be found in the Supporting Information S3.

The principal models used for bioavailability normalization are those that previously applied for the Ni PNEC and EQS setting in Europe (EC, 2008; Nys et al., 2016). The Ni bioavailability models have the following general structure:

$$EC10_{Ni^{2+},t}^{pred\ model\ j} = 10^{-(Q_{species\ i} + S_{pH,model\ j} \times pH_t)} \times (1 + K_{Cat_zBL,model\ j} \times \{Cat^{2+}\}_t) \quad (1)$$

In this model $EC10_{Ni^{2+},t}^{pred,model j}$ is the predicted EC10 or NOEC based on Ni^{2+} activity in target water t (mol/L) based on bioavailability model j ; $K_{Cat,zBL,model j}$ comprises the stability constants for binding of cation z to the biotic ligand for model j (l/mol) and $\{Cat^{2+}\}_t$ the chemical activities of cation z in target water t (mol/L); $Q_{species i}$ is the specific intrinsic sensitivity of species i ($\log_{10}(\text{mol/L})$); $S_{pH,model j}$ is the slope of the log-linear effect of pH on Ni^{2+} toxicity of the chronic Ni bioavailability model j ; and pH_t is the pH of target water t . The structure of the Ni bioavailability models is the same for the four species for which a Ni bioavailability model is available (*P. subcapitata*, *D. magna*, *C. dubia*, and *O. mykiss*), although the number of cations considered differs between the species. Whereas the *P. subcapitata* model only considers a Mg^{2+} competition effect at the biotic ligand site (Deleebeeck et al., 2009a), both the invertebrates and fish models consider a Mg^{2+} competition effect as well as a Ca^{2+} competition effect (Deleebeeck et al., 2008; Deleebeeck, De Schampelaere, & Janssen, 2007; Deleebeeck, Muysen, et al., 2007; De Schampelaere et al., 2006). In addition, the magnitude of the biotic ligand stability constant and the pH slope differs between the models for the different species. To allow application of the full BLM to waters with a pH value between 8.2 and 8.7 and to waters with Ca concentrations between 0.5 and 2 mg L^{-1} the Ni bioavailability models were extended following the approach detailed by Nys et al. (2016) and Peters et al. (2018). All model parameters are available in the Supporting Information S3. Ecotoxicity data that were identified as both relevant and reliable were used for bioavailability normalization when the tests were conducted within the applicability range of the species-specific bioavailability model. The applicability range was defined by the ranges of the key parameters pH and Ca over which the species-specific bioavailability normalization model used for normalization for the considered species could be applied (Supporting Information, Table S2.1), that is, the water chemistry ranges over which the species-specific bioavailability model has been calibrated and validated (compare the definition in EC, 2021). The species-specific bioavailability models were used to normalize the ecotoxicity data for each species, as detailed in the Supporting Information S3. This required the intrinsic sensitivity coefficient (Q) to be calculated from the ecotoxicity test data (Supporting Information S1). The ecotoxicity data for each species were then normalized to the site-specific water chemistry conditions to calculate a site-specific EC10 value for each species as a dissolved Ni concentration under the local water chemistry conditions. These site-specific EC10 values were then used to compile the site-specific SSD from which the site-specific HC5 value can be derived to identify the maximum acceptable dissolved Ni concentration in the local water chemistry conditions.

The bioavailability normalization was performed for a variety of different water chemistries to provide a comparison against the approach used for the European Union RAR (EC, 2008) and EQS (EC, 2011). Seven of these scenarios represent different European ecoregions and were included in the Environmental

and Social Risk (ESR) assessment (EC, 2008). There are also 15 additional scenarios, including a common synthetic water used for ecotoxicity testing (US Environmental Protection Agency moderately hard water), the water chemistry conditions in mesocosm studies that have tested Ni toxicity, waters with a high pH or low Ca concentration that would have been outside the applicability range of the models used in previous regulations, sensitive waters with low DOC concentrations, and example waters from Eastern European countries that were not represented within the ESR ecoregions (Metals Environmental Research Associations, 2020), to increase regional representativity. The water chemistry of each scenario is detailed in the Supporting Information S4. Of the 22 scenarios examined in our study, a detailed comparison of the species averaging approaches was performed for a subset of 11 of these waters, to evaluate differences in the species-specific results used for the SSD calculated by each approach (Figure 2). Collectively, these scenarios illustrate how Ni toxicity varies with changes in water chemistry conditions.

Selection of bioavailability normalization models for cross-species extrapolation

Bioavailability normalization models typically assume that the interactions between Ni and relevant toxicity-modifying factors are the same across related species (see Peters et al., 2018; Schlek et al., 2010). The only variable that needs to be determined when one is applying the model to different species is sensitivity to Ni, and this was performed by calculating a species-specific intrinsic sensitivity coefficient (Q) following the approach used by Peters et al. (2021). For substances like Ni with large and taxonomically diverse ecotoxicity databases, the validity of cross-species extrapolation of bioavailability models is of primary importance in ensuring adequate environmental protection.

The *P. Subcapitata* model was used for the bioavailability normalization of all data for algae (Deleebeeck, De Schampelaere, & Janssen, 2009), and the *O. mykiss* model was used for all data for vertebrates (Deleebeeck, De Schampelaere, & Janssen, 2007). For invertebrates and plants, the existing data were re-analyzed to assess cross-species applicability of bioavailability models as additional data have become available since previous exercises (Deleebeeck, Muysen, et al., 2007; Schlek et al., 2010). Test data for a species were considered sufficient for a bioavailability model comparison when toxicity data were available for a specific endpoint tested in at least two different test media, differing in physicochemistry. Because there are two models available for the bioavailability normalization of invertebrate data (*D. magna* and *C. dubia* bioavailability models) both models were applied, and the best fitting model was identified for each invertebrate taxonomic group, except for the cladocerans. For the cladocerans, each species that could be evaluated was assessed independently because both bioavailability models were developed for cladocerans. For the aquatic plants, three different bioavailability normalization models were considered, the

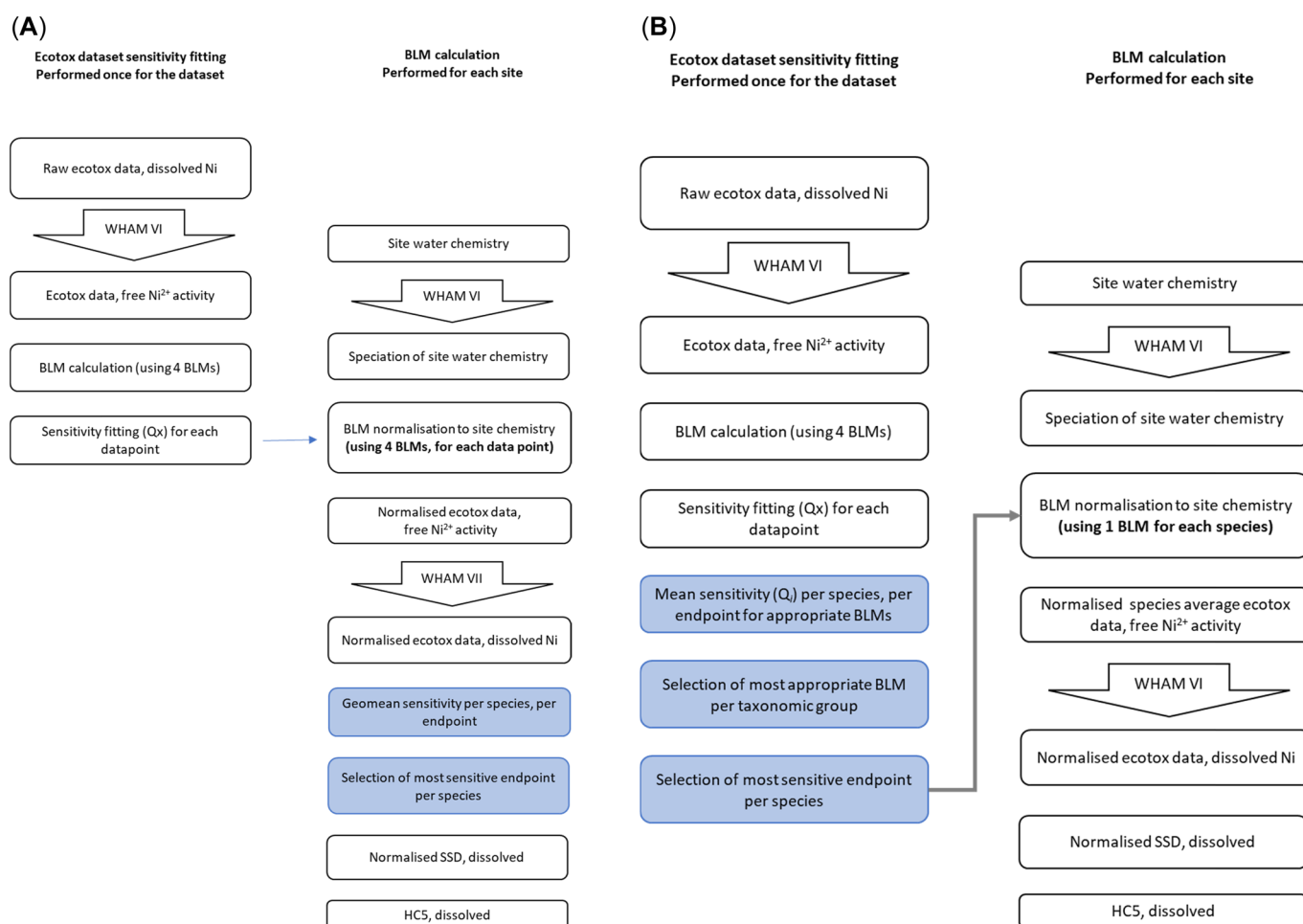


FIGURE 2: Schematic showing the various stages involved in performing the biotic ligand model following species averaging approaches based on averaging of bioavailability normalized dissolved concentrations **(A)** and averaging of species-specific intrinsic sensitivity coefficients **(B)**. HC5 = hazardous concentration to 5% of the species; SSD = species sensitivity distribution; WHAM = Windeermere Humic Aqueous Model.

P. subcapitata model, the *D. magna* model, and the *C. dubia* model, because previous studies have found the sensitivity of these organisms to be better represented by models developed for invertebrates than algae (Peters et al., 2018; Schlegel et al., 2010).

Three approaches were taken for assessing the most appropriate bioavailability model for the normalization of data for each taxonomic group, and this was selected based on the overall weight of evidence from these approaches: 1) The model that resulted in the least variability in the values of the calculated sensitivity coefficients (Q , as identified by the lowest standard deviation in the calculated values) was selected for the normalization when one performed better than the others. 2) The BLM predictions for the original test medium were made for all test results that could be used for the assessment for a species or taxonomic group based on the averaged sensitivity coefficients for the endpoints. The proportion of results that were predicted within a factor of 2, and the overall magnitude of the errors in the BLM-predicted results of the tests were evaluated for all relevant tests for a species or taxonomic group, considering all combinations of test durations and endpoints together. 3) The best fitting

model (as identified by the lowest standard deviation in the calculated Q values) was selected for the most sensitive endpoint for a species or taxonomic group.

Compilation of bioavailability normalized SSDs

For calculation of the site-specific HC5, toxicity data should be aggregated using an SSD. The value taken forward in the SSD calculation is typically the geometric mean of normalized EC10s for the most sensitive endpoint for each species (ECHA, 2008a). Two averaging approaches were explored for compiling the data for the site-specific SSD. The first approach (Figure 2A) was consistent with that used for the existing PNEC and EQS derivation for Ni (EC, 2008; Nys et al., 2016). This approach calculates a site-specific EC10 value, expressed at the dissolved level, for each individual test result before averaging the results for equivalent endpoints (i.e., those based on the same test duration and endpoint). This is implemented using a geometric mean to derive an average value for each species based on the most sensitive endpoint for use in the SSD. The second approach follows the procedure used by Peters et al. (2021) for species averaging (Figure 2B). This

approach uses the arithmetic mean to calculate an intrinsic sensitivity coefficient (Q value) for each species based on the calculated sensitivity coefficient from equivalent endpoints, with the most sensitive endpoint selected for use in the SSD. The use of the arithmetic mean, rather than a geometric mean, is used for the calculation of species-specific intrinsic sensitivity coefficients because this calculation is based on log-transformed data. The calculated Q values for each endpoint are summarized in the Supporting Information S5.

An HC5 was calculated based on the log-normal distribution (ECHA, 2008a), that is, a normal distribution was fitted to the log-transformed EC10 values for each species, expressed at the dissolved level, using the methodology described by Aldenberg et al. (2002). However, several other models have also been identified as being appropriate for SSD fitting, for example, log-logistic, Weibull, gamma, and Gumbel (Fox et al., 2021). In addition, it has been reported that the most commonly used distribution for SSD fitting, the log-normal distribution, failed to fit the distribution of toxicity data in several assessments (Newman et al., 2000). The SSDs were fitted using SSDtools (Thorley & Schwartz, 2018). Therefore, in addition to the normal distribution, a series of alternative distributions was also fitted to the log-transformed data, that is, gamma, logistic, Weibull, Gumbel, and log-normal. The best fitting distribution for each water chemistry scenario was selected based on the Anderson–Darling goodness-of-fit test, which gives more weight to the tails of the SSD (Stephens, 1982).

RESULTS

Data compilation

In total, the newly identified chronic toxicity data for Ni comprised 152 toxicity data points (individual EC10s or NOECs) for 32 species from 28 recent studies that were identified. Compared with the Ni RAR (EC, 2008), which included 214 individual high-quality toxicity data for 31 different freshwater species, chronic toxicity data for an additional 25 species were added: two species of the unicellular alga *Chlorella* sp. (Kakadu and Papua New Guinea strains); the unicellular alga *Naviculla peliculosa*; the amphibians *Lithobates sylvaticus* and *Rhinella arenarum*; the amphipod *Gammarus pseudomnaceus*; the annelids *Lumbriculus variegatus* and *Tubifex tubifex*; the cladocerans *Daphnia ambigua*, *D. galeata*, *D. lumholtzii*, *D. pulex*, and *D. similis*; the fish *Acipenser fulvescens*, *Cottus bairdi*, *Melanotaenia splendida splendida*, and *Oncorhynchus tshawytscha*; the higher plants *Lemna aequinoctialis* and *Spirodela polyrrhiza*; the hydrozoan *Hydra viridissima*; the insects *Chironomus dilutus*, *Hexagenia* sp., and *Neocloeon triangulifer*; the molluscs *Lampsilis siliquoidea*; and finally the rotifer *Euchlanis dilata*. In addition, the EC10 or NOEC values compiled within the Ni RAR data collection effort were checked against the most recent published effect values. This resulted in a correction of 9 and 3 NOECs and EC10s, respectively, from the Ni RAR (Supporting Information S1).

The final freshwater Ni chronic ecotoxicity database (Supporting Information S1) contains 366 individual high-quality chronic ecotoxicity data for 56 different species, belonging to

different taxonomic groups and having different ecologically relevant endpoints (mortality, reproduction, hatching, abnormalities, and growth): algae (10 unicellular green algae species and 1 diatom), amphibians (2 frogs and 3 toads), fish (7 different species, belonging to 5 families), invertebrates (2 amphipods, 2 annelids, 13 cladocera, 2 hydrozoa, 5 insects, 2 rotifers, 3 molluscs), and higher plants (4 species; see the Supporting Information, Table S5.1). An overview of all species included in the database, as well as the endpoints and test durations of the toxicity data entries is given in the Supporting Information S7. The entire database with detailed information including the water chemistry of the tests is given in the Supporting Information S1. An explanation for the exclusion of ecotoxicity data from the final database is provided in the Supporting Information S1.

For the majority of data points, the included EC10 or NOEC values are based on measured data, which are preferred over data based on nominal concentration. However, for three species only nominal effect concentrations could be found: *Hydra littoralis*, *Brachidanio rerio*, and *S. polyrrhiza*, and the data for these species are therefore not included in the final SSD. Although these species are not represented in the chronic database, their removal does not affect overall the taxonomic diversity of the database. The data set includes 222 toxicity data entries obtained in synthetic water; 108, 33, and 3 data entries were obtained in natural surface water, well water, or tap water, respectively. Within the database, pH ranges between 5.5 and 8.8 (median 7.4), hardness between 1.1 and 1098 mg CaCO₃ L⁻¹ (median 100 mg CaCO₃ L⁻¹), and DOC between 0 and 26 mg DOC L⁻¹ (median 0 mg L⁻¹). Non-normalized effect concentrations ranged between 1.1 µg Ni/L (for *C. dubia* and *L. stagnalis*) and 4790 µg Ni L⁻¹ (for *Xenopus laevis*). Comparisons of toxicity data obtained in a diverse set of different test media must consider bioavailability, such as by using the Ni bioavailability models. For six data points, the physicochemistry is outside the boundaries of the species-specific Ni chronic bioavailability models, and they cannot be used to perform bioavailability normalizations. One species (*S. polyrrhiza*) is excluded from the bioavailability normalized SSD based on this criterion. The bioavailability normalization is therefore applied to 358 data points for 53 species.

Selection of bioavailability normalization models

Virtually all the taxonomic groups for which there is potential uncertainty surrounding the most appropriate bioavailability model for normalization (i.e., plants and invertebrates) have sufficient data available for at least one species to allow an assessment to be made of the most appropriate model to use for normalization.

Plants. There were data for two plant species (*L. aequinoctialis* and *L. minor*) covering three endpoints that are suitable for assessing the best fitting bioavailability normalization model, for a total of 17 tests. Variability in the values of the calculated sensitivity coefficients, based on the standard deviation, is summarized in Table 1. This analysis finds that the *C. dubia* model consistently provides the lowest variability, and the

TABLE 1: Summary of model selection and fitted Q values for cladocerans, insects, and molluscs for each endpoint

Taxonomic group	Species	Duration (days)	Endpoint	No. of data	Pseudokirchneriella subcapitata			Daphnia magna			Ceriodaphnia dubia		
					Mean Q	SD		Mean Q	SD		Mean Q	SD	
Plant	<i>Lemna aequinoctialis</i>	4	Growth	4	6.421	0.743		5.135	0.711		1.120	0.557	
Plant	<i>Lemna minor</i>	7d	Root length	10	6.373	0.432		5.199	0.398		0.978	0.354	
Plant	<i>Lemna minor</i>	7	Growth rate	3	6.168	1.160		5.104	0.922		1.088	0.588	
Cladoceran	<i>Ceriodaphnia dubia</i>	7	Reproduction	20	NA	NA		5.519	0.341		1.604	0.221	
Cladoceran	<i>Ceriodaphnia dubia</i>	7	Mortality	4	NA	NA		5.286	0.197		1.308	0.277	
Cladoceran	<i>Ceriodaphnia pulchella</i>	17	Reproduction	2	NA	NA		4.493	0.176		0.711	0.176	
Cladoceran	<i>Ceriodaphnia quadrangula</i>	17	Mortality	4	NA	NA		4.768	0.298		0.987	0.298	
Cladoceran	<i>Ceriodaphnia quadrangula</i>	17	Reproduction	4	NA	NA		4.598	0.282		0.816	0.282	
Cladoceran	<i>Daphnia longispina</i>	21	Mortality	2	NA	NA		4.272	0.118		0.491	0.118	
Cladoceran	<i>Daphnia longispina</i>	21	Reproduction	2	NA	NA		3.804	0.081		0.022	0.081	
Cladoceran	<i>Daphnia lumholzi</i>	21	Survival	2	NA	NA		4.533	0.022		0.725	0.015	
Cladoceran	<i>Daphnia lumholzi</i>	21	Reproduction	2	NA	NA		4.217	0.112		0.410	0.075	
Cladoceran	<i>Daphnia magna</i>	21	Reproduction	31	NA	NA		4.601	0.324		0.779	0.608	
Cladoceran	<i>Daphnia magna</i>	21	Mortality	26	NA	NA		4.369	0.262		0.658	0.419	
Cladoceran	<i>Peracantha truncata</i>	17	Mortality	2	NA	NA		4.326	0.176		0.545	0.176	
Cladoceran	<i>Peracantha truncata</i>	17	Reproduction	2	NA	NA		4.654	0.640		0.873	0.640	
Cladoceran	<i>Simocephalus vetulus</i>	21	Mortality	2	NA	NA		4.503	0.206		0.722	0.206	
Cladoceran	<i>Simocephalus vetulus</i>	21	Reproduction	2	NA	NA		4.503	0.206		0.722	0.206	
Insect	<i>Chironomus dilutus</i>	10	Dry wt	5	NA	NA		3.645	0.326		−0.252	0.436	
Insect	<i>Chironomus tentans</i>	10	Survival	4	NA	NA		3.270	0.506		−0.696	0.436	
Insect	<i>Chironomus tentans</i>	10	Biomass	3	NA	NA		3.451	0.287		−0.558	0.095	
Molluc	<i>Lymnaea stagnalis</i>	30	Wet wt	6	NA	NA		6.175	0.644		2.271	0.674	
Molluc	<i>Lymnaea stagnalis</i>	30	Survival	4	NA	NA		4.516	0.592		0.525	0.438	
Molluc	<i>Lymnaea stagnalis</i>	28	Growth rate (length)	2	NA	NA		5.122	0.489		0.724	0.292	

NA = not available.

P. subcapitata model always performed worst. However, based on the proportion of predictions that were within a factor of 2, both the *C. dubia* and *P. subcapitata* models perform equivalently, at 45%, whereas the *D. magna* model performs worst (40%; Supporting Information S8). The *C. dubia* model also resulted in the lowest mean and maximum prediction errors. The most sensitive endpoint for plants in the chronic Ni database depends on the model used for the normalization, and for the *C. dubia* model this is a single test with *L. gibba*. Overall, the *C. dubia* model is the most appropriate bioavailability normalization model for aquatic plants.

Cladocerans. Two discrete endpoints were tested with *D. magna* covering a total of 57 individual tests. The *D. magna* model results in lower variability than the *C. dubia* model for both endpoints (21-day mortality and 21-day reproduction; Table 1). Based on the bioavailability calculations for the entire *D. magna* data set, the *D. magna* model performs considerably better than the *C. dubia* model, resulting in 74% of predictions within a factor of 2 compared with 42% for the *C. dubia* model, and is therefore used for normalization of *D. magna* data.

Two endpoints were tested for *C. dubia*: 20 tests covering 7-day reproduction and 4 tests covering 7-day mortality (Figure 3C). For the 7-day reproduction endpoint, the *C. dubia* model results in the lowest variability, although the difference between the *C. dubia* and *D. magna* models is less pronounced than it is for the *D. magna* data set. In contrast, the *D. magna* model performs better for the 7-day mortality endpoint. Overall, the *C. dubia* model performs slightly better based on the entire data set, with 83% of results predicted within a factor of 2 compared with 79% for the *D. magna* model. Therefore, this model is used for normalization of the *C. dubia* data. A summary of the calculated sensitivity coefficients is shown in Table 1, and the model fits for each individual test result for *D. magna* and *C. dubia* are shown in the Supporting Information S8.

Two endpoints were available for *D. lumholtzi* for which two test results are available for each. The *C. dubia* model results in lower variability for both endpoints for this species. For all

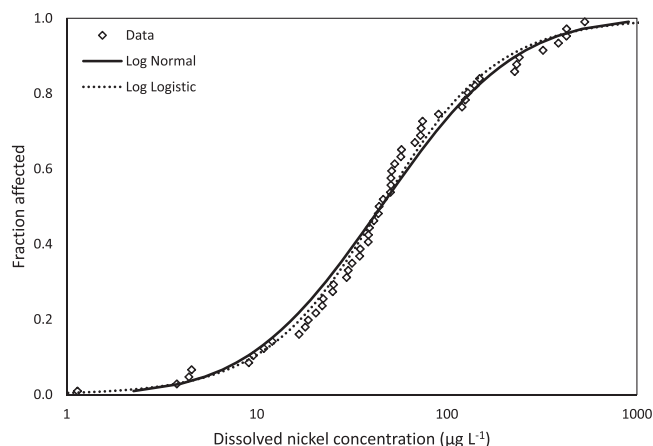


FIGURE 3: Log-normal (solid curve) and log-logistic (dotted curve) distributions fitted to the species sensitivity distributions for Lake Monate.

other cladoceran species for which multiple test results were available for the same endpoint, both the *D. magna* and *C. dubia* models perform equally well, and the *D. magna* model is therefore considered to be an appropriate bioavailability normalization model for these species because it is considered to be more robust in terms of independent validation, which supports the use of this model for these other cladoceran species.

Insects. Both the *D. magna* and *C. dubia* bioavailability models were considered for the insect species. Data were available for two species of chironomid midges covering three different endpoints and including a total of 12 test results (Table 1). The *C. dubia* model provides the lowest variability for two of these endpoints, and the *D. magna* model results in lower variability for one endpoint. Both models result in 58% of predictions being within a factor of 2 of the measured result. When the bioavailability model predictions for all the test results for which model performance can be evaluated were considered together (see Supporting Information S8), the *D. magna* model is seen to provide a marginally better overall fit based on both the mean and maximum prediction errors, whereas the *C. dubia* model provides a slightly better fit when assessed on the factor difference between the experimental and predicted results. The most sensitive endpoint for which model fit can be evaluated was for *C. dilutus* for which the *D. magna* model provides a slightly better fit. Whereas both of the available invertebrate bioavailability normalization models present a relatively similar standard deviation in the intrinsic sensitivity, there was a markedly improved fit of the *C. dubia* model on the three *C. tentans* biomass results. Overall, the *C. dubia* model was the most appropriate bioavailability normalization model for normalization of insect data, although there is uncertainty surrounding this.

Molluscs. Both the *D. magna* and *C. dubia* models were considered for the mollusc species. There were three different endpoints for *L. stagnalis* covering 12 individual test results (Table 1). The *C. dubia* model provides the lower variability for two endpoints, and the *D. magna* model provides marginally lower variability for the other endpoint. When all the relevant data were considered together (see Supporting Information S8) the *C. dubia* model is seen to provide a slightly better fit than the *D. magna* model, and results in a slightly higher proportion of results (57%) within a factor of 2 of the experimental result compared with the *D. magna* model (50%). The most sensitive endpoint for which model performance could be evaluated was for the 30-day wet weight of *L. stagnalis*, for which the *D. magna* model performed marginally better than the *C. dubia* model. Overall, the *C. dubia* model was the most appropriate bioavailability normalization model for normalization of mollusc data, although there is uncertainty surrounding this.

Other invertebrates. The *D. magna* model provided a lower variability to the data for both *H. viridissima* (12 test results) and *B. calyciflorus* (9 test results) than the *C. dubia* model. Consequently the *D. magna* model was applied for the

bioavailability normalization of data for all hydrozoans and rotifers. No data were available for evaluating the best fitting model for either of the worms *L. variegatus* or *T. tubifex*, and the *D. magna* model was used for these species because it provides the best fitting approach for most types of invertebrates.

Summary

Applying the most conservative of the *D. magna* and *C. dubia* bioavailability models without evaluating model performance should be avoided because this will introduce uncertainty into the calculations in an arbitrary way (Peters et al., 2021). In the present study we have demonstrated that the *C. dubia* model should be applied for aquatic plants, insects, molluscs, and the invertebrates *C. dubia* and *D. lumholtzi*, whereas the *D. magna* bioavailability model should be applied for all other cladocerans and invertebrates. There is greater uncertainty surrounding the most appropriate model for those taxa that use the *C. dubia* model. The bioavailability normalization model selected for the critical endpoint for each species is summarized in the Supporting Information S5.

Species sensitivity distributions and HC5 calculations. The water chemistry scenarios used for the site-specific SSDs cover a broad range of pH, DOC, and Ca concentrations to reflect as diverse a range of water chemistry conditions as possible within a limited number of scenarios. The scenarios included cover the seven ecoregions used by the ESR assessment (EC, 2008), three studies on Ni in freshwater mesocosms, waters that would previously have been outside the applicability boundaries of the bioavailability models due to having either high pH or low Ca concentrations, waters that are expected to be sensitive to Ni toxicity, and waters from regions of Europe that were not included in the ESR assessment such as Eastern European countries. The water chemistry conditions covered by the 22 scenarios are pH values between 6.1 and 8.7, DOC concentrations between 0.3 and 16.9 mg L⁻¹, and Ca concentrations between 0.6 and 106 mg L⁻¹. A full summary of the pH, DOC, and Ca concentrations in the scenarios is given in Table 2 along with the HC5 values calculated based on a log-normal distribution and the best fitting distribution. Log-normal HC5 values for the 22 European water scenarios ranged between 1.6 µg/L (Antsoen Finland—soft water conditions) and 35.9 µg/L (Ditches—ecoregion scenario). The best fitting distribution was identified based on the Anderson–Darling statistic. The log–log-normal distribution (i.e., the log-normal distribution fitted to log-transformed toxicity data) was most commonly identified as the best fitting distribution (16 scenarios), and the log-logistic distribution was identified as the best fitting distribution for the remaining scenarios (six scenarios). The results of the distribution fitting are provided in the Supporting Information S9. Best fitting HC5 values for the 22 European water scenarios ranged between 1.7 µg/L (Antsoen, Finland—soft water conditions) and 43 µg/L (Ditches—ecoregion scenario). Differences between the log-normal HC5 and the HC5 based on the best fitting distribution were relatively small, with a maximum difference of 33% for the insensitive Spanish low-pH, high-DOC scenario.

The log-normal and best fitting (log-logistic) SSDs are both shown for the Lake Monate scenario in Figure 3.

Comparisons were performed between the two approaches for deriving a single value for each species to use in the SSD on a subset of the water chemistry scenarios. These were the ESR ecoregions and other waters to cover the extremes of the range of water chemistry conditions covered by the scenarios. The two approaches toward selecting the values to be used in the SSD for each species, that is, averaging at the sensitivity coefficient level or at the dissolved concentration level, produced very similar results for the majority of species in the SSD. Species-specific predictions for all species following both species averaging approaches are within 5% of each other in all the water chemistry scenarios considered. However, some minor differences between the two approaches were observed, although these were dependent on both the species and the water chemistry scenario used for normalization. The greatest differences were observed for *L. stagnalis*, *L. minor*, *L. aequinoctialis*, and *M. splendida splendida* (see Figure 4); notably, these are all species with relatively high levels of variability between different test results for the critical endpoints. Species with limited variability in sensitivity between different tests, or only a single ecotoxicity test result for the critical endpoint, showed no differences in the two approaches. Consequently, the approach based on averaging sensitivity coefficients (Q values) to derive an intrinsic sensitivity coefficient for each species was selected as the approach for calculating site-specific SSDs for all the water chemistry scenarios because this approach is considerably less calculation intensive.

The bioavailability normalization procedure in the present study includes updates compared with the previous approach in terms of the data set, the cross-species extrapolation, and the level on which species averaging takes place. The effects of updates in the bioavailability normalization procedure on the HC5 calculations are explored in more detail in the Supporting Information S6. In the comparison, the HC5 values reported in the Ni RAR (EC, 2008) are used as the basis for comparison. Correcting some of the NOEC/EC10 values in the Ni RAR data set lowered the calculated HC5 with by an 0.89-fold median value relative to the values reported in the Ni RAR data set. As a next step of updating the procedure, the cross-species extrapolation was updated using the recommendations reported in the bioavailability model selection above. This update lowered the HC5 compared with the Ni RAR–reported values by 0.94-fold (median). Finally, the newly identified high-quality toxicity data were added to the database. This had a clear effect on the normalized HC5 calculations, with HC5 being on average approximately 22% (median) lower than those reported in the Ni RAR. The largest effect on the HC5 was noted by incorporating additional data for the species that were already present in the database, whereas the effects of adding new species to the database were more moderate.

DISCUSSION

The present study updated the existing chronic Ni effects assessment method for Ni by incorporating newly published

TABLE 2: Summary of water chemistry scenarios and HC5 values

Name	Code	pH	DOC (mg L ⁻¹)	Ca (mg L ⁻¹)	Log-normal HC5 (µg L ⁻¹)	Lower 90% CL	Upper 90% CL	Best fit distribution	Best fit HC5	Lower 90% CL	Upper 90% CL	Log-normal Anderson–Darling statistic	Best fit Anderson–Darling statistic
Lake Monate Rhine	ESR 1 ESR 2	7.7 7.8	2.5 2.8	13.6 68.9	5.61 8.61	3.54 4.9	9.28 15.7	Log-log istic Log-log- normal	5.78 8.34	3.22 NA	9.91 NA	0.43 0.695	0.304 0.195
Otter Teme	ESR 3 ESR 4	8.1 7.6	3.2 8	46.9 50.1	6.94 15	3.8 9.21	12.6 25.5	Log-logistic Log-log- normal	7.32 15.2	3.95 NA	13.3 NA	0.584 0.725	0.432 0.175
Swedish lake within boundary Ebro	ESR 5 ESR 6	6.7 8.2	3.8 3.7	8.7 72.9	9.48 7.63	5.95 4.17	16.3 14.3	Log-log- normal Log-log- normal	11 6.81	7.05 NA	18.2 NA	0.743 0.627	0.433 0.148
Ditches adjusted DOC + H	ESR 7	6.9	12	60.1	35.9	21.3	63.9	Log-log- normal	43	27.6	70.9	0.955	0.551
Homen et al. 2016 lower	Mesocosm 1	8.7	3.4	62.6	2.85	1.53	5.36	Log-log- normal	2.7	NA	NA	0.535	0.135
Nys et al., 2019 lower	Mesocosm 2	8.3	16	14	19	12.2	30.3	Log-log- normal	14.1	NA	NA	0.492	0.187
Mebane et al., 2020	Mesocosm 3	7.7	3	5	6.1	3.9	9.98	Log-logistic	6.33	3.89	10.9	0.272	0.16
USEPA moderately hard water	Synthetic test water	7.6	0.3	14	4.25	2.27	7.79	Log-logistic	4.46	2.35	8.54	0.427	0.29
Austria	High pH 1	8.6	1.2	68.3	2.7	1.41	5.49	Log-log- normal	2.51	NA	NA	0.486	0.124
Spain	High pH 2	8.5	2.2	37.2	2.76	1.58	5.38	Log-log- normal	2.51	NA	NA	0.408	0.136
France soft	Soft water 1	6.4	3.1	0.6	2.58	1.62	4.29	Log-log- normal	2.41	NA	NA	0.288	0.164
Atnsjöen Finland	Soft water 2	6.1	1.5	0.8	1.57	0.926	2.8	Log-log- normal	1.65	NA	NA	0.427	0.194
River Duddon/ Derwent	Sensitive water 1	6.5	0.7	2	5	2.95	8.84	Log-log- normal	5.67	NA	NA	0.505	0.319
River Bourne Hampshire	Sensitive water 2	8	0.4	106	6.09	3.32	12.5	Log-log- normal	5.85	NA	NA	0.683	0.209
France sensitive	Sensitive water 3	8.3	0.6	35.8	2.44	1.29	4.72	Log-logistic	2.66	1.29	5.29	0.519	0.349
Spain	Low pH, high DOC	6.3	8.5	51	25.5	14.8	47.2	Log-log- normal	33.8	22.1	56	0.94	0.417
Romania	Eastern Europe 1	7.8	2.9	38.4	7.03	4.14	12.7	Log-log- normal	6.46	NA	NA	0.636	0.176
Latvia	Eastern Europe 2	7.9	16.9	55.5	23.6	15.1	38.9	Log-log- normal	21.9	NA	NA	0.695	0.166
France additional	Moderate pH low Ca	7.2	2.2	5.8	5.53	3.52	9.55	Log-logistic	5.49	3.15	9.31	0.375	0.289

Also given are the HC5 values calculated based on a log-normal distribution and the best fitting distribution. CL = confidence limit; DOC = dissolved organic carbon; HC5 = hazardous concentration for 5% of a species; NA = not available; USEPA = US Environmental Protection Agency.

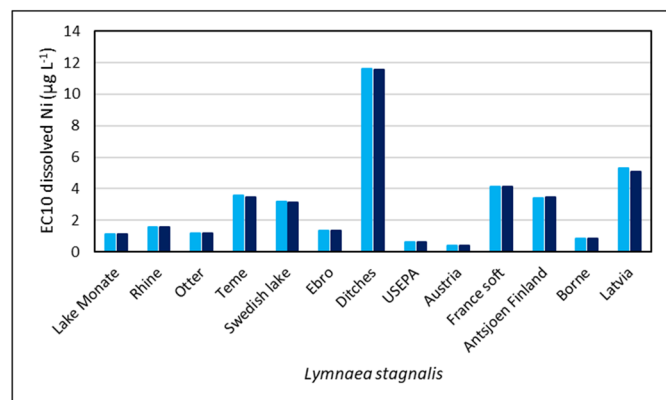
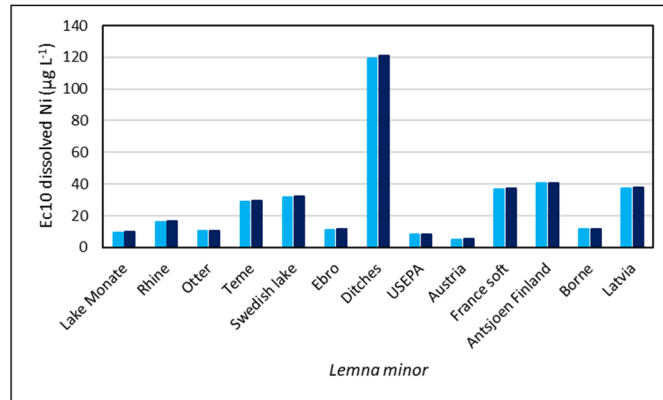
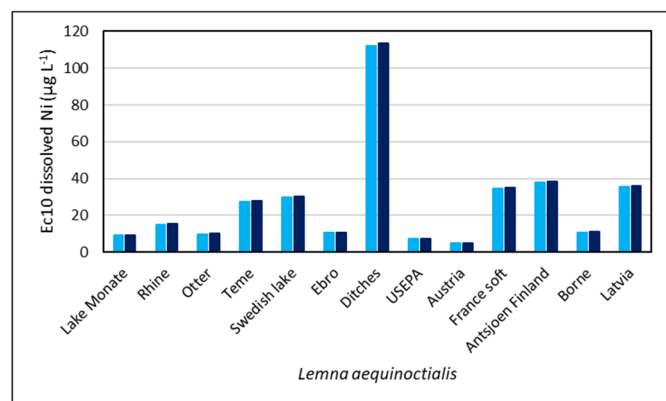
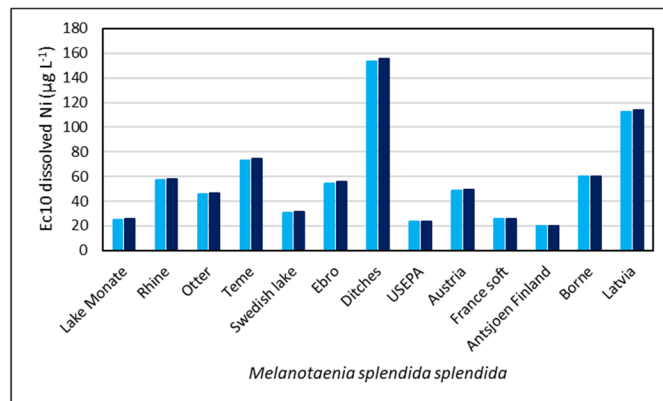
(A) *Lymnaea stagnalis*(C) *Lemna minor*(B) *Lemna aequinoctialis*(D) *Melanotaenia splendida splendida*

FIGURE 4: Comparison of species averaging approaches (light blue, intrinsic sensitivity coefficients; dark blue, bioavailability normalized dissolved concentrations) used for species sensitivity distribution compilations for those species showing the greatest differences between the two approaches. (A) *Lymnaea stagnalis*. (B) *Lemna aequinoctialis*. (C) *Lemna minor*. (D) *Melanotaenia splendida splendida*. EC10, 10% effective concentration.

ecotoxicity data and integrating more recent modifications of the existing chronic Ni bioavailability models. This updated Ni effects assessment method can be used to update regulatory hazard assessments for Ni in Europe, such as the Ni EQS under the WFD and hazard classifications, and for deriving a PNEC under REACH. A total of 53 species of 10 taxonomic groups are represented within the database, which satisfies the minimum taxonomic representation requirements for the use of an SSD (ECHA, 2008a) of at least 10 species covering a diverse range of taxonomic groups. The present study also covers a broader range of water chemistry conditions, and given that both high pH and low Ca represent higher Ni bioavailability, the incorporation of these modifications into the bioavailability normalization approach for Ni means that the Ni full BLM can be applied to more sensitive conditions. Approximately 20% of the European surface waters have a pH higher than 8.2 and/or a Ca concentration lower than 2 mg/L, based on reported conditions in the FOREGS data set (Salminen et al., 2005). Incorporating these updates in the full BLM normalization procedure allows an additional 18% of these waters to be included.

The approach taken toward the selection of a single value for each species for use in the SSD differs from that taken

previously (EC, 2008; Nys et al., 2016). The calculations in the present study are based on a species-specific intrinsic sensitivity coefficient, rather than calculating a bioavailability normalized result for every single ecotoxicity test in the database before averaging. Although it has not previously been applied in Europe, this approach has been used for bioavailability normalization in Australia (Peters et al., 2021) and offers several advantages over the historical approach. The initial benefit is that the level of variability in organism response can be expressed based on the variability in the intrinsic sensitivity coefficients, which aids in accounting for intra- and interlaboratory variations and differences between different clones or strains of a species. This approach allows the performance of different bioavailability models to be evaluated based on their ability to reduce variation in the calculated sensitivity coefficients for a species. This also reduces the number of speciation calculations required for each specific set of water chemistry conditions to the number of species in the ecotoxicity database ($n = 55$), rather than the number of individual tests ($n = 360$). Although this approach provides a straightforward means to compare the sensitivities of different species or endpoints independently of the water chemistry conditions, it does not provide bioavailability normalized predictions for any

endpoints other than the critical (i.e., most sensitive) endpoint for each species. The intrinsic sensitivity approach also results in some very minor differences in the predicted sensitivity of species with a high level of variability in their intrinsic sensitivities compared with averaging for each species on the basis of bioavailability normalized dissolved concentrations (see Figure 4).

There is a relatively high degree of variability in test results for some species, which has implications for evaluating bioavailability model performance and for the selection of the best fitting model for the taxonomic group and can also result in differences between the two species averaging approaches. This is particularly notable for the aquatic plants and the mollusc *L. stagnalis*. Tests on aquatic plants appear to exhibit a relatively high degree of variability, as noted previously by Peters et al. (2018, 2020). The data set for *L. minor* covers four different studies and three different endpoints. Although an Organisation for Economic Co-operation and Development guideline (2016) is available for chronic tests on *L. stagnalis*, none of the tests follow that guideline method. Virtually all tests with *L. stagnalis* are juvenile growth tests and follow several different methods. The importance of other confounding factors influencing Ni sensitivity to *L. stagnalis* such as test duration, food sources, snail holding conditions, and snail age at test initiation has been suggested (Cremazy et al., 2020) and may account for at least some of the variability among these test results.

The application of species-specific bioavailability models to species other than those that they were originally developed for is well established (Deleebeeck, De Schampelaere, & Janssen, 2007; Deleebeeck, Muysen, et al., 2007; Deleebeeck, De Laender, et al., 2009; Peters et al., 2018; Schlekot et al., 2010). These studies have consistently found that bioavailability models for invertebrates rather than unicellular algae are better at normalizing data for aquatic plants, and this is assumed to be due to both Ca and Mg being relevant as competing ions, because only Mg is considered in the algal bioavailability normalization model. There are also two different species-specific bioavailability normalization models for invertebrates, and identifying which of these results in the greatest reduction in uncertainty for each taxonomic group is necessary. The difference between these two models is in the magnitude of the effect of pH on Ni sensitivity, as indicated by the slopes of the pH coefficients used by the models (see Supporting Information S3). This means that the pH range over which species have been tested is likely to be important for identifying which model performs best. However, except for *D. magna* and *C. dubia*, most species for which the most appropriate bioavailability normalization model has been evaluated were assessed for a specific test duration and endpoint and cover only a relatively limited range of pH values. For example, there are six results for *L. stagnalis* based on 30-day wet weight, and these tests cover a pH range between 6.9 and 8.0.

The selection of the most appropriate bioavailability model for the normalization of toxicity data for taxonomic groups follows an approach different from that taken previously in PNEC and EQS setting (Nys et al., 2016; Schlekot et al., 2010).

This modification was made because there are two different bioavailability normalization models that have been developed for invertebrate species (i.e., a *D. magna* model and a *C. dubia* model). The previous approach that was used for the Ni PNEC and EQS derivation in Europe determined the most appropriate bioavailability normalization model for plant or invertebrate species for which sufficient data existed for that species (EC, 2008). At the time, there were fewer species represented in the database, and several taxonomic groups that did not have sufficient data to allow an assessment to be performed. The approach taken was to perform calculations based on both invertebrate bioavailability normalization models and select the lowest normalized EC10, that is, the most sensitive, result for use in calculating the SSD. Concerns have been raised that the approach of using the bioavailability model that provides the most sensitive result introduces additional uncertainty to the assessment (Peters et al., 2021).

The approach followed for the present study does not rely on a single analysis, but rather takes account of several different approaches toward evaluating which model provides the best bioavailability normalization for different species and taxonomic groups. The best bioavailability normalization model is taken to be that which results in the least variability in the calculated sensitivity coefficients and the bioavailability normalized EC10 values. In some cases, the best fitting model is clear, although this tends to be for taxa in which the *D. magna* model provides the best bioavailability normalization. In other cases, especially for species and taxa that exhibit a greater level of variability in the data, the model that provides the best bioavailability normalization is somewhat equivocal. This is especially true for the insects and molluscs, both of which use the *C. dubia* model for normalization of data for all species from these taxonomic groups. Previous assessments, such as Schlekot et al. (2010) and Peters et al. (2021), have relied on a single assessment approach. Schlekot et al. (2010) used model predictions of test results, and Peters et al. (2021) used variability in calculated species sensitivity coefficients. The present study combines both approaches to ensure that the conclusions drawn are robust.

Santore et al. (2021) suggested that the high sensitivity of *C. dubia* to Ni toxicity at high pH may be due to bicarbonate toxicity. Both *N. triangularis* (Besser et al., 2021) and *L. siligoidea* (Harper et al., 2014) have also been identified as sensitive to bicarbonate. The findings of the analysis of the best fitting bioavailability normalization model for both of these species, namely, that the *C. dubia* model is the most appropriate model for both insects and molluscs, concurs with these findings, although in both cases this is based on analyses for other species.

A potential limitation is the way in which the full BLM is applied to soft waters, because the sensitivity coefficients are based on the same ecotoxicity database as that used for higher hardness waters but calculated using the updated model constants. The ecotoxicity data set includes a large quantity of data that was collected under much higher hardness conditions than are covered by the version of the full BLM that has been adapted for application to soft waters. Although this means

that the data set used for each of the model applicability ranges is consistent, it also introduces the uncertainty that much of the ecotoxicity data used are not appropriate for the applicability range of the model. Given that the competitive effect of Ca and Mg on Ni toxicity is greater in very soft waters, the inclusion of ecotoxicity data from tests in hard waters is likely to distort the sensitivity coefficient toward higher sensitivity. This factor is likely to make predictions made using the soft water version of the BLM more protective than they would be for exclusively soft water species. Applying the soft water bioavailability models only to toxicity data collected in soft waters would be more appropriate but would result in a very different and much smaller data set than is applied otherwise. Updating the bioavailability models to an empirical approach that incorporates the changes in Ca and Mg binding affinity

with changes in the water hardness, as noted by Peters et al. (2018), would improve this situation by enabling the complete range of water hardness to be covered by a single model.

Derivation of HC5 values for water chemistry scenarios

The HC5s were calculated for 22 freshwater scenarios. In the European surface water scenarios, log-normal HC5 values ranged between 1.6 µg/L (Antsjöen, Finland) and 36 µg/L (Ditches). These water scenarios serve as examples of the range of HC5 values that can result from the updated bioavailability normalization procedure for Ni. The range of bioavailability normalized HC5 values observed in the present study clearly shows the importance of incorporating bioavailability when performing

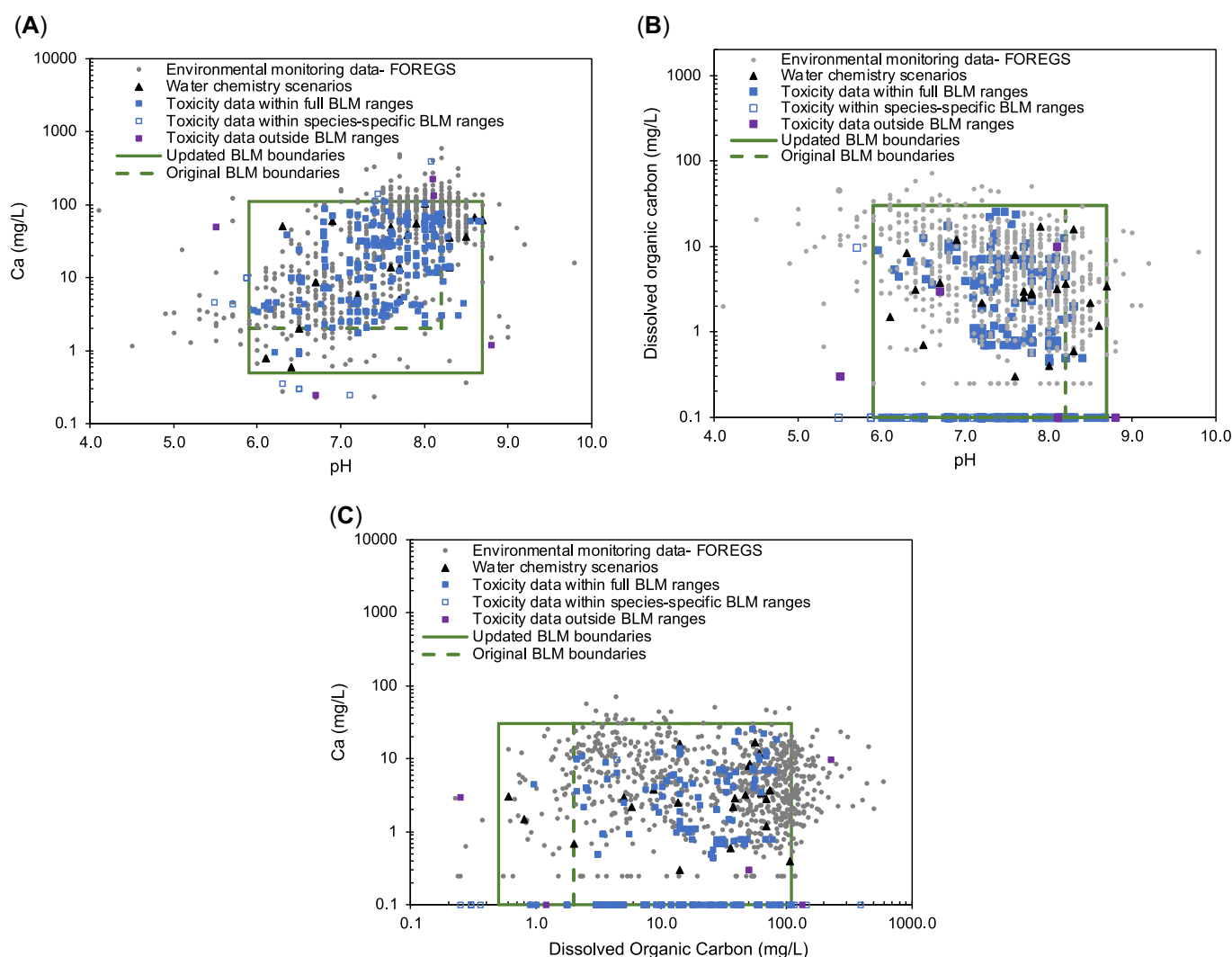


FIGURE 5: Overview of water chemistry relationships represented as pH versus Ca concentrations (A), pH versus dissolved organic carbon (DOC) concentrations (B), and DOC versus Ca concentrations (C) in European surface waters (small gray circles, as reported in the FOREGS database; Salminen et al., 2005), in test media included in the toxicity database (squares) and water chemistry scenarios used for bioavailability normalization (black triangles). For the chemistry of the test media of the toxicity data, solid blue squares indicate data points within the full biotic ligand model (BLM) limits; open squares indicate data points outside the full BLM limits, but within the species bioavailability model selected for normalization; purple squares are data points outside both the full BLM limits and the species bioavailability model selected for normalization. Green lines indicate the boundaries of the full BLM; solid lines represent those of the updated full BLM (present study); and dashed lines represent those of the original full BLM (EC, 2008). FOREGS, Forum of European Geological Survey.

site-specific compliance checking. Some of the waters selected represent extreme water chemistry conditions (e.g., Antsjöen, Finland) compared with the overall range of surface waters in Europe (Figure 5). Limited differences were generally observed between the log-normal HC5 and the best fitting distribution. The log-log-normal distribution was most commonly identified as the best fitting distribution, although in several cases confidence intervals could not be calculated for the HC5 value. For this reason, and for consistency with the existing approach, the log-normal distribution is recommended as an appropriate distribution, although in many cases the log-logistic distribution provided a slightly better fit (see Supporting Information S9). The best fitting distribution did not provide HC5 values that were consistently either higher or lower than the log-normal distribution, and in most cases both HC5 values are closely comparable.

The HC5 values calculated in the present study are protective of the effects observed in both mesocosm (Hommen et al. 2016; Mebane et al., 2020; Nys et al., 2019) and field-based research (Peters et al., 2014) and indicate that the bioavailability normalization approach is protective for all taxa, including early life stages of insects (Mebane et al., 2020). The lowest NOEC from Hommen et al. (2016) for the most sensitive taxon, *L. stagnalis*, was at least two times higher than the bioavailability normalized HC5 for that water chemistry. Nys et al. (2019) also found no clear effects on community structure and functioning endpoints at a concentration higher than the HC5 calculated for the water chemistry. The most sensitive endpoint derived by Mebane et al. (2020), based on a 50% decline in the abundance of the mayfly *Diphetor hageni*, was more than three times higher than the HC5 value calculated for that water chemistry. Collectively, these studies suggest that the HC5 values derived in the present study are protective of ecosystems overall under a variety of different water chemistry conditions.

CONCLUSIONS

The ecotoxicity database for chronic Ni toxicity to freshwater organisms has been updated and now includes 360 individual data points covering a total of 55 different species, all of which are suitable for bioavailability normalization to different water chemistry conditions. The bioavailability normalization procedure has also been updated to include updates to the bioavailability models that enable more sensitive water chemistry conditions to be covered by the model predictions. The updated database and bioavailability normalization procedure are applicable to more than 95% of regulated European surface water conditions. This updated database and bioavailability normalization procedure provide a scientifically robust basis for the derivation of regulatory thresholds (such as PNEC and EQS values) for chronic Ni toxicity in freshwaters and are protective of ecosystem-level effects evaluated based on multiple mesocosm studies. The present update serves to further strengthen the already very strong data set and models.

Supporting Information—The Supporting Information is available on the Wiley Online Library, at <https://doi.org/10.1002/etc.5561>.

Acknowledgment—The present study was funded by NiPERA.

Disclaimer—The authors work for, or as subcontractors to, NiPERA.

Author Contributions Statement—**Adam Peters**: Conceptualization; Methodology; Investigation; Formal analysis; Writing—original draft; Project administration; Writing—review & editing. **Charlotte Nys**: Methodology; Investigation; Formal analysis; Writing—original draft. **Dean Leverett**: Formal analysis; Writing—review & editing. **Iain Wilson**: Resources; Data curation; Writing—review & editing. **Patrick Van Sprang**: Writing—review & editing. **Graham Merrington**: Supervision; Writing—review & editing. **Elizabeth Middleton**: Supervision; Project administration; Writing—review & editing. **Emily Garman**: Writing—review & editing. **Christian Schlegat**: Supervision; Writing—review & editing.

Data Availability Statement—All the data included in our study are either already published, or are included in the Supporting Information.

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